Chapter 2

Literature Review

This chapter reviews the work on which HepatoTwin builds and the gaps it addresses. It follows the structure of the system: the clinical problem and the evidence for dietary treatment (Section 2.1); how liver fat is measured without biopsy (Section 2.2); machine-learning models that estimate steatosis from routine data, and the limits on how well they can do so (Section 2.3); how diet is measured and personalised (Section 2.4); digital twins in health and nutrition (Section 2.5); computational gastronomy and food recommendation (Section 2.6); and how a twin can be evaluated when no longitudinal data are available (Section 2.7). Section 2.8 summarises the gap.

2.1 MASLD and Its Dietary Management

2.1.1 Definition and burden

Metabolic dysfunction-associated steatotic liver disease (MASLD) is defined as hepatic steatosis together with at least one of five cardiometabolic risk factors, in the absence of another cause [1]. The 2023 multi-society nomenclature replaced non-alcoholic fatty liver disease (NAFLD), removed the reliance on exclusion and stigmatising terms, and introduced MetALD for patients with MASLD who also drink moderate amounts of alcohol. A meta-analysis of studies using imaging to diagnose NAFLD estimated a global adult prevalence of 30.05%, rising from 25.26% in 1990–2006 to 38.00% in 2016–2019 [2]; a second meta-analysis reached a similar overall estimate [3]. Most patients have simple steatosis, but a minority progress to steatohepatitis, fibrosis and cirrhosis, and the disease is closely associated with type 2 diabetes and cardiovascular risk [4, 5].

2.1.2 Evidence for lifestyle and dietary treatment

Both the European (EASL-EASD-EASO) and American (AASLD) guidelines name lifestyle change as the first-line treatment for most patients [4, 5]. The strongest evidence concerns weight loss. In a prospective study of 293 patients with paired liver biopsies, weight loss through lifestyle change produced dose-dependent histological improvement: losses of 5% or more improved steatosis and inflammation, and losses of 10% or more resolved steatohepatitis in 90% of patients and led to fibrosis regression in 45% [6]. A meta-analysis of 22 randomised trials with 2,588 participants found that weight-loss interventions lowered ALT by 9.81 U/L and improved liver fat and histological activity, but found no significant change in fibrosis [7]. The thresholds of 5%, 7–10% and 10% weight loss are now written into the European guideline [4]. Liver fat also responds quickly: in people with type 2 diabetes on an 800 kcal per day diet, hepatic triglyceride fell from 12.8% to 2.9% over eight weeks, and had already fallen by 30% after the first week [8].

The evidence for specific dietary components is more mixed, and this variation matters for any system that turns evidence into advice. A 2026 meta-analysis of randomised trials, which graded the certainty of each finding with GRADE, found that a Mediterranean-style diet reduced MRI-PDFF by 1.37 percentage points and improved liver stiffness, while omega-3 supplementation and low-carbohydrate high-fat diets did not change ALT [9]. Dietary interventions including calorie restriction improved liver enzymes, steatosis and stiffness in another meta-analysis [10]. Yet a meta-analysis of ten trials found that the Mediterranean diet reduced AST and GGT but not ALT [11], and it was no better than a low-fat diet in the short term [12].

Feeding trials show that the type of surplus energy matters. When 38 overweight adults were overfed by 1,000 kcal per day for three weeks, liver fat rose by 55% with saturated fat, 33% with simple sugars and 15% with unsaturated fat [13]. A meta-analysis of 13 trials with 260 participants found that fructose raised liver fat and ALT when it added surplus energy, but had no effect when it replaced other carbohydrate in the same amount of energy [14]. Exercise also reduces liver fat: in a meta-analysis of trials measuring liver fat by MRI, exercise was associated with a relative reduction of 30% or more, the threshold that predicts histological response, even without clinically significant weight loss [15].

Two lessons follow. First, the evidence is graded: biopsy-based dose–response data for weight loss are stronger than meta-analyses of liver enzymes, and null findings sit alongside positive ones. A system that recommends food should therefore carry the strength of the evidence with each recommendation, rather than treating all claims as equal. Second, the evidence is expressed in terms of energy, weight change, nutrients

and dietary patterns, never single recipes; any mapping from evidence to a recipe is an inference that must be made explicit.

2.2 Non-Invasive Assessment of Liver Fat

Liver biopsy is the reference standard for grading steatosis and fibrosis but is invasive and impractical for screening. Magnetic resonance imaging proton-density fat fraction (MRI-PDFF) and proton spectroscopy (^{1}H-MRS) measure liver fat accurately but are costly. Vibration-controlled transient elastography (FibroScan) is cheaper and widely used: its controlled attenuation parameter (CAP), reported from 100 to 400 dB/m, grades steatosis, with cut-offs of 248, 268 and 280 dB/m for grades S1, S2 and S3 derived from an individual patient data meta-analysis [16].

CAP is, however, an imperfect measure of liver fat. A review of the technique reports that agreement between two operators examining the same patient has a concordance correlation of 0.82, falling to 0.65 in patients with BMI of 30 kg/m^2 or more and to 0.44 when CAP is below 240 dB/m [17]. The same review notes that CAP accuracy falls when the interquartile range (IQR) of the ten acquisitions exceeds 40 dB/m, which is why that limit is used as a quality criterion. In adults with NAFLD, CAP correlated with MRI-PDFF at $r = 0.66$, falling to $r = 0.37$ for changes over time [18]. Part of every CAP reading is therefore measurement variation, and the variation is largest in exactly the people who are hardest to examine. This limits how closely any model can predict CAP (Section 2.3.3).

Simple clinical scores offer a cheaper alternative. The Fatty Liver Index (FLI) combines BMI, waist circumference, triglycerides and GGT [19]; the Hepatic Steatosis Index (HSI) combines the ALT/AST ratio, BMI, sex and diabetes [20]; and the NAFLD liver fat score combines metabolic syndrome, type 2 diabetes, fasting insulin, AST and the AST/ALT ratio, reaching an AUC of 0.86–0.87 against spectroscopy, with genetic information adding less than 1% [21]. These scores are the natural baselines for any new steatosis model.

2.3 Machine-Learning Prediction of Steatosis and Liver Fat

2.3.1 Classification of steatosis in NHANES

The US National Health and Nutrition Examination Survey (NHANES) added FibroScan to its 2017–March 2020 cycle, and several studies have since trained classifiers to detect CAP-defined steatosis from routine data (Table 2.1). Gradient-boosted trees, usually XGBoost [22], perform best or near best in most of them, and the

strongest predictors are consistently body-size measures (waist circumference and BMI), followed by liver enzymes, glucose and insulin markers, and triglycerides. Model explanations, typically SHAP values [23], are widely reported.

Du et al. used 6,814 participants and found logistic regression, XGBoost and LightGBM comparable (AUC 0.815–0.842), so that the added complexity of boosting bought little [24]. Liu et al. compared models on an automated machine-learning platform in 4,177 participants; XGBoost reached an AUC of 0.859, and the authors imputed missing values by multiple imputation and reported calibration curves and decision-curve analysis [25]. Zhang et al. reported an AUC of 0.874 on 2,460 complete cases [26]. Boullion et al. identified biomarkers of steatosis and stiffness; propensity matching on age and sex required complete data and reduced the steatosis sample to 2,714, so their models were trained on the larger unmatched data, where random forests reached 79.6% accuracy [27]. Zhu et al. trained a three-grade model on 215,145 non-obese Chinese adults, reaching a macro-average AUC of 0.941 internally and 0.874 in 726 non-Hispanic Asian NHANES participants with complete data [28]; this is one of the few studies with external validation, and it shows the usual fall in performance when a model moves to a new population.

Table 2.1: Machine-learning classifiers of CAP-defined steatosis using NHANES.

Study	**Population**	**Outcome**	**Best model**	**Performance**
Du et al. (2026) [24]	6,814 adults, 2017–2020	MASLD, CAP ≥ 280	LR, XGBoost, LightGBM	AUC 0.815–0.842
Liu et al. (2024) [25]	4,177, fasting subsample; imputed	CAP ≥ 302	XGBoost (AutoML)	AUC 0.859; calibration curve
Zhang et al. (2025) [26]	2,460 complete cases	CAP-defined MASLD	XGBoost	AUC 0.874
Boullion et al. (2025) [27]	Unmatched sample (matched subset 2,714)	CAP ≥ 238	Random forest	Accuracy 79.6%
Zhu et al. (2026) [28]	215,145 non-obese Chinese adults; 726 NHANES Asians	CAP grades (S0, S1, S2–S3)	XGBoost	Macro AUC 0.941 internal; 0.874 external

Three limitations recur. Most studies use only the 2017–2020 cycle, and most discard participants with any missing value. Calibration is reported only occasionally [25], although a risk model whose output will be shown to patients must produce probabilities that mean what they say; isotonic regression is a standard way to obtain them from tree ensembles [29]. And all are single snapshots, used to classify a participant once rather than to track or support them over time.

2.3.2 Prediction of liver fat as a continuous value

Fewer studies predict the amount of liver fat, and their results are more sobering (Table 2.2). In 28,796 UK Biobank adults, the Fatty Liver Index explained 28% of the variance in MRI-PDFF (33–37% across cross-validation folds); an extended model with twelve variables (FLI+) explained 54% on average across four cross-validation folds (0.520–0.557) [30]. A UK Biobank preprint reported R^2 of 0.37 from anthropometric and blood measures in 40,533 participants with MRI-PDFF and 0.34 from plasma metabolites in 9,723, against 0.63–0.64 from DXA body-composition images analysed by a convolutional network [31]. Haas et al., also in UK Biobank, found that BMI correlated with MRI liver fat at r = 0.42 and that adding 24 clinical factors and biomarkers raised the correlation only to r = 0.58 ($R^2 \approx 0.34$), concluding that liver fat cannot be reliably estimated from clinical factors [32]. In Finnish adults measured by spectroscopy, liver enzymes alone explained 15–25% of the variance; an equation including a specialist marker (phosphorylated IGFBP-1) reached an adjusted R^2 of 0.49 in validation and an AUROC of 0.84 for NAFLD, against 0.72 for FLI and 0.62 for HSI [33]. Higher values come from small, selected samples: in 272 patients of a Chinese obesity clinic, a four-variable equation predicted CAP with R^2 = 0.76 in derivation and 0.65 in validation [34], and a four-variable index correlated with MRI-PDFF at ρ = 0.74 in 149 overweight or obese adults, validated only by bootstrap resampling [35]. No study reporting R^2 for CAP in NHANES was found.

Table 2.2: Prediction of liver fat as a continuous value.

Study	Population	Reference	Inputs	R^2
UK Biobank preprint (2024) [31]	40,533; 9,723; about 46,000 UK adults	MRI-PDFF	Body and blood measures; metabolites; DXA images	0.37; 0.34; 0.63–0.64
Asaturyan et al. (2022) [30]	28,796 UK adults	MRI-PDFF	FLI; FLI with 12 variables (FLI+)	0.28–0.36; 0.54 (4-fold mean)
Haas et al. (2021) [32]	UK adults, 1,214 test	MRI	BMI; BMI with 24 clinical factors	*r* 0.42; *r* 0.58
Petäjä et al. (2016) [33]	378 Finnish adults	^{1}H-MRS	Enzymes only; equation with phosphorylated IGFBP-1	0.15–0.25; 0.49 (adjusted)
Liu et al. (2022) [34]	272, obesity clinic	CAP	BMI, neck circumference, TG, C-peptide	0.76 / 0.65
Chen et al. (2023) [35]	149, overweight or obese	MRI-PDFF	ALT, AST, insulin, 1-h glucose	$\rho = 0.74$
Kotronen et al. (2009) [21]	470 Finnish adults	^{1}H-MRS	Metabolic and liver markers	Not reported

2.3.3 Why prediction from routine data is bounded

Two findings explain why continuous liver fat is hard to predict from routine measurements. First, liver fat is strongly heritable: in a prospective study of monozygotic and dizygotic twins using MRI-PDFF, the heritability of steatosis was 0.52 after adjustment for age, sex and ethnicity [36], and in UK Biobank measured genetic variants together explained up to 30% of the variance in liver fat [32]. Variants such as those in PNPLA3 change liver fat at the same body size, and routine clinical data contain none of this information. Second, the reference measurement itself is noisy (Section 2.2): a model can explain no more of the variation in a single CAP reading than the reading shares with the patient's true liver state. Together, these findings imply that a model built on routine data should reach a moderate R^2 for liver fat, between about 0.3 and 0.55 in the large studies above, and good discrimination of steatosis, which is the pattern seen in Tables 2.1 and 2.2. For a digital twin, the implication is that a calibrated probability of steatosis is a more defensible state variable than a single predicted CAP value.

2.4 Dietary Assessment and Personalised Nutrition

Personalising dietary advice requires knowing what a person usually eats. NHANES collects one or two 24-hour dietary recalls per participant. A single recall describes the average intake of a group well, but an individual's usual intake poorly, because intake varies from day to day. Statistical methods developed at the US National Cancer Institute estimate usual intake from two or more recalls by separating within-person from between-person variation [37]. For foods that are not eaten every day, the NCI method uses a two-part model that separates the probability of eating a food on a given day from the amount eaten on days it is eaten, allows the two parts to be correlated, and can add covariates such as age, sex or a food frequency questionnaire; in simulation it estimated the distribution of usual intake substantially better than earlier methods [38]. Self-reported intake is also biased: in the OPEN study, comparison with recovery biomarkers showed substantial under-reporting, particularly of energy [39]. Diet quality can be summarised with the Healthy Eating Index-2020, a 0–100 score of 13 components aligned with the 2020–2025 Dietary Guidelines for Americans; its components are unchanged from HEI-2015 for ages 2 and over, and a separate index was introduced for toddlers [40]. Computing it requires food-group data from the USDA Food Patterns Equivalents Database.

Personalisation is motivated by individual variation in response. Zeevi et al. showed in 800 people that post-meal glucose responses to identical meals differ widely and can be predicted from personal data, and that diets personalised on these predictions lowered post-meal glucose [41]. For a liver twin, these findings suggest two requirements: dietary inputs should represent usual rather than single-day intake, and the system should become more personal as more of the patient's own data arrive.

2.5 Digital Twins in Health and Nutrition

2.5.1 Definitions and requirements

The US National Academies define a digital twin as a set of virtual information constructs that mimics the structure, context and behaviour of a physical system, is dynamically updated with data from it, has a predictive capability and informs decisions; the bidirectional flow between the virtual and physical system is central, and credibility rests on verification, validation and uncertainty quantification (VVUQ) [42]. A scoping review of health twins frames them as a physical entity, its virtual representation and the connection between them, and found that the term is used for systems ranging from static models to fully coupled ones [43]. A perspective on medical twins describes them as most advanced in cardiology and oncology, and names data integration, validation and privacy as the main obstacles [44].

Two architectural proposals are relevant to a modular twin. Nitschke et al.\ propose a general patient twin built on a bipartite knowledge graph that links patient attributes to existing predictive models; fusion models based on ensemble learning combine the models' outputs, and a propagation scheme passes information through the graph each time new patient data arrive. They describe the design as modular, evolving, informed, interpretable and explainable [45]. Nye structures patient data as a knowledge graph built on medical ontologies such as SNOMED CT and proposes closed-form continuous-time ("liquid") neural networks to model the patient's evolving state [46]. Both are conceptual designs without empirical evaluation, but they share a principle that HepatoTwin adopts: separate modules, each with a defined role, connected by an explicit patient state.

2.5.2 Metabolic and nutrition twins

In nutrition, Gkouskou et al. extended the idea of an "actual digital twin", in which a patient is matched to many similar patients whose outcomes are known, into a "virtual digital twin": a personal model combining genetic, longitudinal metabolomic, immune, microbial and behavioural data, collected through connected devices and managed by machine learning, to simulate dietary interventions for that person [47]. Metabolic twins have also been outlined as tools that track a person's metabolic flexibility over time [48]. A systematic review of twins in precision health identified 12 studies and found improvements in most of the reported outcomes, while noting that the evidence is still small and heterogeneous [49].

The most developed clinical example is the Twin Health platform. Its method paper describes a machine-learning model (CatBoost) that predicts each person's post-meal glucose response from continuous glucose monitoring, food logs, activity and sleep, and a mobile application that adjusts dietary recommendations from those predictions and keeps refining them as data arrive [50]. In a one-year randomised trial in 319 people with type 2 diabetes, nutrition guided by this twin improved liver-fat scores and MRI-PDFF compared with standard care [51]. A second example, iDT-diet, modelled how diet quality affects biomarkers over time in 402 participants measured at baseline and after one, three and six years. It combined a random forest with Choquet-integral LASSO feature selection, which had lower error than XGBoost, LightGBM and CatBoost for several targets, with generated natural-language descriptions and three-dimensional avatars of the predicted body state [52]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [53].

These systems either recommend individual foods from predicted glucose responses or forecast biomarker trajectories. The Twin Health platform depends on continuous glucose monitoring and proprietary data, and iDT-diet forecasts biomarkers without recommending food. None evaluates complete, existing recipes against evidence-graded dietary priorities, and none combines health with cuisine and practicality when making a recommendation.

2.5.3 From prediction to "what if"

A twin that compares interventions must answer counterfactual questions, which predictive accuracy alone does not license. Freeman et al. name two traps. The *fidelity trap* is the belief that a model accurate enough to reproduce patient trajectories can therefore say what would happen under a different treatment; prediction and counterfactual estimation are distinct tasks with different justifications. The *feedback trap* arises when a twin learns from data that its own recommendations helped to produce, which breaks the assumptions behind its causal estimates. They argue that health twins should be causally valid, modular and evolving systems, with modularity isolating the parts that need interventional justification [54]. Laudy formalises a twin as a simulator of each person's potential outcomes under treatment and control, and proposes a hierarchy of fidelity (marginal, joint and structural) with a five-level validation regime that turns the claim "the simulator produces correct counterfactuals" into tests against observable data, while stating that the dependence between a person's two potential outcomes remains untestable [55].

For HepatoTwin, these arguments have a direct consequence. A cross-sectional model trained on observational data, such as a steatosis classifier built on NHANES, describes the current state well but cannot by itself predict the effect of changing a diet. Intervention effects must instead come from interventional evidence, held in a separate module with their strength and uncertainty made explicit, and the twin's updating must be evaluated in a setting where the true effect is known (Section 2.7).

2.6 Computational Gastronomy and Food Recommendation

2.6.1 Recipe and food resources

Computational gastronomy treats food and cooking as structured data, studying recipes through their flavour, nutrition, health associations and sustainability [56]. RecipeDB contains 118,171 recipes from six continents, 26 geocultural regions and 74 countries, cooked with 268 processes from over 20,000 ingredients, with cuisine labels, utensils and nutritional profiles, and ingredients linked to their flavour molecules [57]. FlavorDB

catalogues 25,595 flavour molecules, 2,254 of which are linked to 936 natural ingredients in 34 categories [58]. Literature mining of 5,769 abstracts produced 8,957 associations between 152 culinary spices and 848 diseases; for 150 of the 152 spices the beneficial associations outnumbered the adverse ones, and the authors used this balance to make disease-specific spice recommendations [59]. These resources make it possible to reason about recipes at scale, but they are built for exploration and analysis rather than for deciding which recipes suit a particular patient.

2.6.2 Food recommender systems

Food recommendation faces challenges beyond those of other domains: it must balance taste, health, culture and practicality. In a survey of the field, Trattner and Elsweiler showed that re-weighting standard recommender outputs by WHO or UK Food Standards Agency health scores can trade accuracy for healthiness, but the healthiest lists came with very poor accuracy, and they named health-aware recommendation an open problem [60].

Knowledge graphs are a common foundation. FoodKG links over a million recipes, their ingredients and USDA nutrient data in 67 million triples, and supports queries such as which recipe can be made from the ingredients at hand while avoiding allergens [61]. Chen et al. framed personalised recommendation as constrained question answering over FoodKG, treating the user's query, dietary preferences and health guidelines as constraints, and improved on non-personalised baselines by 59.7 percentage points on average across metrics [62]. Graph neural networks have been used to move users gradually towards healthier recipes while respecting their preferences [63]. In all of these systems, "healthy" is defined by generic nutritional guidelines rather than by a patient's clinical state, and the evidence behind each recommendation is not graded.

2.6.3 Recipe modification and adaptation

A related line of work modifies recipes by substituting ingredients. Shirai et al. combined explicit knowledge from a food knowledge graph with word embeddings into a substitutability heuristic (DIISH), filtered the candidates by nutritional and food-category constraints, and evaluated it against substitutions taken from cooking guides and recipe reviews [64]. Loesch et al. learned knowledge-graph embeddings (RDF2Vec) over a food graph and evaluated them against 1,847 known substitution pairs, retrieving about 40% of the true substitutes in the top ten candidates [65]. Rita et al. fine-tuned large language models to choose substitutions that raise phytochemical content, improving Hit@1 from 34.5% to 38.0% on the GISMo benchmark and producing 1,639 enriched recipes, while cautioning that the health claims rest on preclinical evidence

[66]. A survey of the area concludes that a good substitute must match flavour, culinary function and health suitability together, and recommends knowledge graphs and neuro-symbolic reasoning for doing so [67]. Cultural adaptation goes further: translating recipes between Chinese and English cuisines requires changing ingredients, units, tools and techniques, and GPT-4 still fell short of human experts when adapting English recipes into Chinese [68].

Every modification creates a new recipe whose taste, feasibility and nutrition must be predicted, and none of these studies tested whether people liked or cooked the result. Evaluating existing recipes, the approach taken in this thesis, avoids that uncertainty: every recommended recipe has already been written, cooked and nutritionally described.

2.6.4 Menu planning, diversity and additional criteria

Menu planning composes meals that satisfy nutritional and cost targets; Sahin et al. released EvoMeal, an open-source tool comparing four multi-objective evolutionary algorithms, and found that two newer algorithms (AGE-MOEA and SMS-EMOA) outperformed NSGA-II and NSGA-III [69]. Recommendation lists also benefit from diversity: maximal marginal relevance re-ranks items by a linear combination of relevance and dissimilarity to the items already chosen, reducing redundancy [70]. Two further criteria are relevant to MASLD guidance. The degree of food processing can be estimated with FPro, a machine-learning score that predicts processing from nutrient profiles; it classified over 73% of the US food supply as ultra-processed, was associated in NHANES with metabolic syndrome and diabetes, and showed that swapping foods for less-processed alternatives could reduce these associations [71]. Glycaemic load is available from the international glycaemic index tables [72].

2.7 Evaluating Digital Twins Without Longitudinal Data

A twin's defining features, updating with new data and projecting the future, can only be fully validated against repeated measurements of the same people. When such data are unavailable, simulation offers a recognised alternative. In-silico clinical trials test interventions on computer-simulated patients before, or alongside, real trials [73]. The best-known example is the UVA/Padova type 1 diabetes simulator, a population of virtual patients built from physiological models, which the US Food and Drug Administration accepted in 2008 as a substitute for animal trials in the preclinical testing of closed-loop insulin control [74, 75]. Its authors stress that such testing is a prerequisite to clinical trials, not a substitute for them [74]. Synthea generates synthetic patient records for software testing at scale [76].

Two designs are particularly relevant. In plasmode simulation, real data provide the covariates and the outcome is simulated from a known model, so the data keep their real structure while the true effect is known and a method's ability to recover it can be measured [77]. In data assimilation, observing system simulation experiments generate a "true" state with one model, give a forecasting system only noisy observations of it, and measure how well the system recovers the truth; the field calls these twin experiments [78].

Simulated follow-up is credible only if its ingredients come from evidence. For diet-driven weight change, Hall et al. developed and validated a dynamic model of how a sustained change in energy intake translates into weight over time: a lasting reduction of about 10 kcal per day eventually lowers weight by about 0.45 kg, with half of the change reached in about one year and 95% in about three years. The common static rule of 3,500 kcal per pound overestimates long-term weight loss by about 100%, and the model predicts average rather than individual responses [79]. Large lifestyle trials report the weight trajectories that real participants achieve. In Look AHEAD, intensive lifestyle intervention produced a mean weight loss of 8.6% at one year against 0.6% with usual care; 68.0% and 37.7% of participants lost at least 5% and 10%, and at eight years the mean losses were 4.7% and 2.1% [80]. The Diabetes Prevention Program reported a similar pattern of loss and partial regain [81], and a Look AHEAD analysis showed that losses of 5–10% increased the odds of clinically meaningful improvements in glycaemia, blood pressure and lipids [82]. The response of the liver itself is given by the weight-loss studies of Section 2.1 [7, 6, 8]. Realistic observations also require the normal biological variation of each laboratory test [83] and the known under-reporting of food intake [39].

These sources make it possible to evaluate a twin's updating on virtual patients whose follow-up is grounded in published human data. Such an evaluation verifies that the twin behaves correctly when the truth is known; it does not establish its accuracy for real patients, which still requires longitudinal data.

2.8 Summary and Research Gap

Table 2.3 summarises what each strand of the literature offers and what it leaves open. Clinical research provides graded evidence on how weight and diet change the liver, but expressed in energy, nutrients and patterns rather than recipes. Machine-learning studies estimate steatosis well from routine data, but as one-off classifications, rarely calibrated, and with a ceiling on continuous prediction set by genetics and measurement. Digital twins in nutrition personalise food choices or forecast biomarkers, but do not evaluate existing recipes against evidence-graded priorities, and the causal literature warns that

predictive accuracy alone does not justify their intervention advice. Food recommender and substitution systems handle preferences and generic healthiness, but not a patient's clinical state or the strength of evidence. And no liver twin has been evaluated for how it updates over time.

Table 2.3: Strands of related work, what they leave open, and how HepatoTwin addresses it.

Strand	**What it provides**	**What it leaves open**	**HepatoTwin**
Clinical evidence (Section 2.1)	Graded effects of weight loss, diet patterns and nutrients	Not expressed for recipes; evidence strength varies	Evidence engine with explicit strength labels, mapped to recipe criteria
Steatosis models (Section 2.3)	Good discrimination from routine data	Snapshots; rarely calibrated; bounded R^2	Calibrated probabilities as the twin's state, within a defined scope
Dietary assessment (Section 2.4)	Recall data; usual-intake methods	Single-day recalls used directly	Recall-adjusted priorities; usual intake updated from food logs
Digital twins (Section 2.5)	Personal state, glucose-response prediction, trajectories	No recipe-level, evidence-graded recommendation; fidelity and feedback traps	Modular twin linking state, evidence and recipes; effects from interventional evidence
Food recommendation (Section 2.6)	Recipe data, preferences, generic healthiness, substitution	No clinical state; ungraded evidence; untested modified recipes	Evidence-gated ranking of existing recipes by health, cuisine and feasibility
In-silico evaluation (Section 2.7)	Virtual patients, plasmode and twin experiments	Not yet applied to liver twins	In-silico evaluation of the twin's updating on virtual NHANES patients

This thesis addresses the intersection of these strands: a patient-specific liver twin whose calibrated state, combined with graded intervention evidence and the patient's own diet, drives the transparent evaluation and ranking of existing recipes, and whose updating is evaluated in silico before real longitudinal data exist.

References

1. Rinella, M. E., Lazarus, J. V., Ratziu, V., Francque, S. M., Sanyal, A. J., Kanwal, F., et al. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology*, 78(6):1966–1986, 2023.
2. Younossi, Z. M., Golabi, P., Paik, J. M., Henry, A., Van Dongen, C. and Henry, L. The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): a systematic review. *Hepatology*, 77(4):1335–1347, 2023. doi:10.1097/HEP.0000000000000004.
3. Riazi, K., Azhari, H., Charette, J. H., Underwood, F. E., King, J. A., Afshar, E. E., Swain, M. G., et al. The prevalence and incidence of NAFLD worldwide: a systematic review and meta-analysis. *The Lancet Gastroenterology & Hepatology*, 7(9):851–861, 2022. doi:10.1016/S2468-1253(22)00165-0.
4. European Association for the Study of the Liver (EASL), European Association for the Study of Diabetes (EASD) and European Association for the Study of Obesity (EASO). EASL-EASD-EASO Clinical Practice Guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD). *Journal of Hepatology*, 81(3):492–542, 2024. doi:10.1016/j.jhep.2024.04.031.
5. Rinella, M. E., Neuschwander-Tetri, B. A., Siddiqui, M. S., Abdelmalek, M. F., et al. AASLD Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease. *Hepatology*, 77(5):1797–1835, 2023. doi:10.1097/HEP.0000000000000323.
6. Vilar-Gomez, E., et al. Weight loss through lifestyle modification significantly reduces features of nonalcoholic steatohepatitis. *Gastroenterology*, 149(2):367–378, 2015. doi:10.1053/j.gastro.2015.04.005.
7. Koutoukidis, D. A., Astbury, N. M., Tudor, K. E., Morris, E., Henry, J. A., et al. Association of weight loss interventions with changes in biomarkers of nonalcoholic fatty liver disease: a systematic review and meta-analysis. *JAMA Internal Medicine*, 179(9):1262–1271, 2019.
8. Lim, E. L., Hollingsworth, K. G., Aribisala, B. S., Chen, M. J., Mathers, J. C. and Taylor, R. Reversal of type 2 diabetes: normalisation of beta cell function in association with decreased pancreas and liver triacylglycerol. *Diabetologia*, 54(10):2506–2514, 2011.
9. Stern, U. M., Wagenpfeil, G., Weber, S. N., Alqahtani, S. A., Michel, M., Schattenberg, J. M., et al. The impact of dietary interventions on liver health biomarkers in individuals with metabolic dysfunction-associated steatotic liver disease: a systematic review and meta-analysis of randomized controlled trials. *European Journal of Nutrition*, 65(2):58, 2026. doi:10.1007/s00394-025-03870-z.
10. Haigh, L., Kirk, C., El Gendy, K., Gallacher, J., Errington, L., et al. The effectiveness and acceptability of Mediterranean diet and calorie restriction in non-alcoholic fatty liver disease (NAFLD): a systematic review and meta-analysis. *Clinical Nutrition*, 41(9):1913–1931, 2022. doi:10.1016/j.clnu.2022.06.037.

11. Sangouni, A. A., et al. Effect of Mediterranean diet on liver enzymes: a systematic review and meta-analysis of randomised controlled trials. *British Journal of Nutrition*, 128(7):1231–1239, 2022. doi:10.1017/S0007114521002270.

12. Xiong, Y., et al. Effects of Mediterranean diet and low-fat diet on liver enzymes and liver fat content of NAFLD: a meta-analysis of randomized controlled trials. *Food & Function*, 15(16):8248–8257, 2024. doi:10.1039/D4FO01461H.

13. Luukkonen, P. K., et al. Saturated fat is more metabolically harmful for the human liver than unsaturated fat or simple sugars. *Diabetes Care*, 41(8):1732–1739, 2018. doi:10.2337/dc18-0071.

14. Chiu, S., et al. Effect of fructose on markers of non-alcoholic fatty liver disease (NAFLD): a systematic review and meta-analysis of controlled feeding trials. *European Journal of Clinical Nutrition*, 68(4):416–423, 2014. doi:10.1038/ejcn.2014.8.

15. Stine, J. G., et al. Exercise training is associated with treatment response in liver fat content by magnetic resonance imaging independent of clinically significant body weight loss in patients with nonalcoholic fatty liver disease: a systematic review and meta-analysis. *American Journal of Gastroenterology*, 118(7):1204–1213, 2023. doi:10.14309/ajg.0000000000002098.

16. Karlas, T., Petroff, D., Sasso, M., Fan, J., Mi, Y., de Lédinghen, V., Kumar, M., et al. Individual patient data meta-analysis of controlled attenuation parameter (CAP) technology for assessing steatosis. *Journal of Hepatology*, 66(5):1022–1030, 2017. doi:10.1016/j.jhep.2016.12.022.

17. Ferraioli, G. Quantitative assessment of liver steatosis using ultrasound controlled attenuation parameter (Echosens). *Journal of Medical Ultrasonics*, 48:489–495, 2021. doi:10.1007/s10396-021-01106-1.

18. Ahn, S. B., Jun, D. W., Kang, B., et al. Optimal cutoff value for assessing changes in intrahepatic fat amount by using the controlled attenuation parameter in a longitudinal setting. *Medicine*, 97(50):e13636, 2018. doi:10.1097/MD.0000000000013636.

19. Bedogni, G., et al. The Fatty Liver Index: a simple and accurate predictor of hepatic steatosis in the general population. *BMC Gastroenterology*, 6:33, 2006.

20. Lee, J., et al. Hepatic steatosis index: a simple screening tool reflecting nonalcoholic fatty liver disease. *Digestive and Liver Disease*, 42(7):503–508, 2010.

21. Kotronen, A., Peltonen, M., Hakkarainen, A., Sevastianova, K., Bergholm, R., et al. Prediction of non-alcoholic fatty liver disease and liver fat using metabolic and genetic factors. *Gastroenterology*, 137(3):865–872, 2009. doi:10.1053/j.gastro.2009.06.005.

22. Chen, T. and Guestrin, C. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

23. Lundberg, S. M. and Lee, S. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, 2017.

24. Du, S., Yu, H., Du, J. and Xu, Y. Machine learning models to predict metabolic dysfunction-associated steatotic liver disease (MASLD) with simple anthropometric and biochemical variables: a cross-sectional study in US population. *International Journal of Hepatology*, 2026:8221645, 2026. doi:10.1155/ijh/8221645.

25. Liu, L., Lin, J., Liu, L., Gao, J., Xu, G., Yin, M., Liu, X., Wu, A. and Zhu, J. Automated machine learning models for nonalcoholic fatty liver disease assessed by controlled attenuation parameter from the NHANES 2017–2020. *Digital Health*, 10:20552076241272535, 2024.

26. Zhang, Y., Liu, X., Zhang, X., Fei, Y. and Li, X. Machine learning-based prediction of metabolic dysfunction-associated steatotic liver disease using National Health and Nutrition Examination Survey (NHANES) data. *PLoS ONE*, 20(11):e0335656, 2025. doi:10.1371/journal.pone.0335656.

27. Boullion, J., Husein, A., Agrawal, A., et al. Machine learning–based biomarker identification for early diagnosis of metabolic dysfunction–associated steatotic liver disease. *The Journal of Clinical Endocrinology & Metabolism*, 110(11):e3866–e3877, 2025. doi:10.1210/clinem/dgaf111.

28. Zhu, Y., Wang, Y., Zhang, S., Yang, J., Zhang, F., Liu, Y., et al. Hepatic steatosis severity prediction in nonobese individuals: machine learning model development and validation. *Journal of Medical Internet Research*, 28:e82529, 2026.

29. Zadrozny, B. and Elkan, C. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.

30. Asaturyan, H. A., Basty, N., Thanaj, M., Whitcher, B., Thomas, E. L. and Bell, J. D. Improving the accuracy of fatty liver index to reflect liver fat content with predictive regression modelling. *PLoS ONE*, 17(9):e0273171, 2022. doi:10.1371/journal.pone.0273171.

31. Machine learning across multiple imaging and biomarker modalities in the UK Biobank improves genetic discovery for liver fat accumulation. medRxiv preprint, 2024. doi:10.1101/2024.01.06.24300923.

32. Haas, M. E., Pirruccello, J. P., Friedman, S. N., et al. Machine learning enables new insights into genetic contributions to liver fat accumulation. *Cell Genomics*, 1(3):100066, 2021. doi:10.1016/j.xgen.2021.100066.

33. Petäjä, E. M., Zhou, Y., Havana, M., Hakkarainen, A., Lundbom, N., Ihalainen, J. and Yki-Järvinen, H. Phosphorylated IGFBP-1 as a non-invasive predictor of liver fat in NAFLD. *Scientific Reports*, 6:24740, 2016. doi:10.1038/srep24740.

34. Liu, H., Li, X., Han, X., et al. Simple surrogate equations to predict controlled attenuation parameter values for screening non-alcoholic fatty liver disease in a Chinese population. *Frontiers in Medicine*, 9:894895, 2022. doi:10.3389/fmed.2022.894895.

35. Chen, J., Yue, J., Fu, J., et al. A prediction model of liver fat fraction and presence of non-alcoholic fatty liver disease (NAFLD) among patients with overweight or obesity. *Endocrine Journal*, 70:977–985, 2023. doi:10.1507/endocrj.EJ23-0227.

36. Loomba, R., Schork, N., Chen, C., et al. Heritability of hepatic fibrosis and steatosis based on a prospective twin study. *Gastroenterology*, 149(7):1784–1793, 2015. doi:10.1053/j.gastro.2015.08.011.

37. Dodd, K. W., Guenther, P. M., Freedman, L. S., Subar, A. F., Kipnis, V., Midthune, D., et al. Statistical methods for estimating usual intake of nutrients and foods: a review of the

theory. *Journal of the American Dietetic Association*, 106(10):1640–1650, 2006. doi:10.1016/j.jada.2006.07.011.

38. Tooze, J. A., Midthune, D., Dodd, K. W., Freedman, L. S., Krebs-Smith, S. M., Subar, A. F., et al. A new statistical method for estimating the usual intake of episodically consumed foods with application to their distribution. *Journal of the American Dietetic Association*, 106(10):1575–1587, 2006. doi:10.1016/j.jada.2006.07.003.

39. Subar, A. F., Kipnis, V., Troiano, R. P., Midthune, D., Schoeller, D. A., Bingham, S., et al. Using intake biomarkers to evaluate the extent of dietary misreporting in a large sample of adults: the OPEN study. *American Journal of Epidemiology*, 158(1):1–13, 2003.

40. Shams-White, M. M., Pannucci, T. E., Lerman, J. L., Herrick, K. A., et al. Healthy Eating Index-2020: review and update process to reflect the Dietary Guidelines for Americans, 2020–2025. *Journal of the Academy of Nutrition and Dietetics*, 123(9):1280–1288, 2023. doi:10.1016/j.jand.2023.05.015.

41. Zeevi, D., Korem, T., Zmora, N., Israeli, D., Rothschild, D., Weinberger, A., Ben-Yacov, O., et al. Personalized Nutrition by Prediction of Glycemic Responses. *Cell*, 163(5):1079–1094, 2015. doi:10.1016/j.cell.2015.11.001.

42. National Academies of Sciences, Engineering, and Medicine. Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press, Washington, DC, 2024. doi:10.17226/26894.

43. Katsoulakis, E., Wang, Q., Wu, H., Shahriyari, L., Fletcher, R., Liu, J., Achenie, L., Liu, H., et al. Digital twins for health: a scoping review. *npj Digital Medicine*, 7(1):77, 2024. doi:10.1038/s41746-024-01073-0.

44. Laubenbacher, R., Mehrad, B., Shmulevich, I. and Trayanova, N. Digital twins in medicine. *Nature Computational Science*, 4(3):184–191, 2024. doi:10.1038/s43588-024-00607-6.

45. Nitschke, A., Brandl, C., Egersdörfer, F., Görtz, M., Hohenfellner, M. and Weidemüller, M. Design for a digital twin in clinical patient care. *npj Health Systems*, 3:11, 2026. doi:10.1038/s44401-025-00060-1.

46. Nye, L. Digital twins for patient care via knowledge graphs and closed-form continuous-time liquid neural networks. arXiv preprint arXiv:2307.04772, 2023.

47. Gkouskou, K., Vlastos, I., Karkalousos, P., Chaniotis, D., Sanoudou, D. and Eliopoulos, A. G. The ``Virtual Digital Twins'' Concept in Precision Nutrition. *Advances in Nutrition*, 11(6):1405–1413, 2020. doi:10.1093/advances/nmaa089.

48. Su, C., Wang, P., Foo, N. and Ho, D. Optimizing metabolic health with digital twins. *npj Aging*, 11(1):20, 2025. doi:10.1038/s41514-025-00211-6.

49. Shen, M. D., Chen, S. B. and Ding, X. D. The effectiveness of digital twins in promoting precision health across the entire population: a systematic review. *npj Digital Medicine*, 7(1):145, 2024. doi:10.1038/s41746-024-01146-0.

50. Shamanna, P., Joshi, S., Thajudeen, M., Shah, L., Poon, T., Mohamed, M. and Mohammed, J. Personalized nutrition in type 2 diabetes remission: application of digital twin technology for predictive glycemic control. *Frontiers in Endocrinology*, 15:1485464, 2024. doi:10.3389/fendo.2024.1485464.

51. Joshi, S., Shamanna, P., Dharmalingam, M., Vadavi, A., Keshavamurthy, A., Shah, L., et al. Digital Twin-Enabled Personalized Nutrition Improves Metabolic Dysfunction-Associated Fatty Liver Disease in Type 2 Diabetes: Results of a 1-Year Randomized Controlled Study. *Endocrine Practice*, 29(12):960–970, 2023. doi:10.1016/j.eprac.2023.08.016.

52. Nobel, A., Matos, J., Wang, H. and Fang, H. iDT-diet: An intelligent digital twin model for diet-influenced biomarker trajectories (Student Abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 41334–41336, 2026. doi:10.1609/aaai.v40i48.42260.

53. ARTEMIS: AcceleRating the Translation of virtual twins towards a pErsonalised Management of steatotIc liver patientS. Horizon Europe grant 101136299; ClinicalTrials.gov NCT07430501, 2024.

54. Freeman, N. L. B., Zou, Y., Lee, K., Yu, C., et al. The fidelity and feedback traps: the case for health digital twins as modular evolving causal systems. arXiv preprint arXiv:2607.27192, 2026.

55. Laudy, O. The digital twin counterfactual framework: a validation architecture for simulated potential outcomes. arXiv preprint arXiv:2604.01325, 2026.

56. Bagler, G. and Goel, M. Computational gastronomy: capturing culinary creativity by making food computable. *npj Systems Biology and Applications*, 10(1):72, 2024. doi:10.1038/s41540-024-00399-5.

57. Batra, D., Diwan, N., Upadhyay, U., Kalra, J. S., Sharma, T., Sharma, A. K., Khanna, D., Marwah, J. S., Kalathil, S., Singh, N., Tuwani, R. and Bagler, G. RecipeDB: a resource for exploring recipes. *Database*, 2020:baaa077, 2020. doi:10.1093/database/baaa077.

58. Garg, N., Sethupathy, A., Tuwani, R., NK, R., Dokania, S., Iyer, A., Gupta, A., et al. and Bagler, G. FlavorDB: a database of flavor molecules. *Nucleic Acids Research*, 46(D1):D1210–D1216, 2018. doi:10.1093/nar/gkx957.

59. Rakhi, N. K., Tuwani, R., Mukherjee, J. and Bagler, G. Data-driven analysis of biomedical literature suggests broad-spectrum benefits of culinary herbs and spices. *PLoS ONE*, 13(5):e0198030, 2018. doi:10.1371/journal.pone.0198030.

60. Trattner, C. and Elsweiler, D. Food recommender systems: important contributions, challenges and future research directions. arXiv preprint arXiv:1711.02760, 2017.

61. Haussmann, S., Seneviratne, O., Chen, Y., et al. FoodKG: a semantics-driven knowledge graph for food recommendation. In *The Semantic Web – ISWC 2019*, 2019.

62. Chen, Y., Subburathinam, A., Chen, C. and Zaki, M. J. Personalized food recommendation as constrained question answering over a large-scale food knowledge graph. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining (WSDM)*, 2021.

63. Li, D., Zaki, M. J. and Chen, C. Health-guided recipe recommendation over knowledge graphs. *Journal of Web Semantics*:100743, 2022. doi:10.1016/j.websem.2022.100743.

64. Shirai, S. S., Seneviratne, O., Gordon, M. E., Chen, C. and McGuinness, D. L. Identifying ingredient substitutions using a knowledge graph of food. *Frontiers in Artificial Intelligence*, 3:621766, 2021. doi:10.3389/frai.2020.621766.

65. Loesch, J., Meeckers, L., van Lier, I., de Boer, A., Dumontier, M. and Celebi, R. Automated Identification of Food Substitutions Using Knowledge Graph Embeddings. In

CEUR Workshop Proceedings, pages 19–28, 2022.

66. Rita, L., Southern, J., Laponogov, I., Higgins, K., et al. Optimizing ingredient substitution using large language models to enhance phytochemical content in recipes. arXiv preprint arXiv:2409.08792, 2024.

67. Kim, H., Venkataramanan, R. and Sheth, A. A survey on food ingredient substitutions. arXiv preprint arXiv:2501.01958, 2025.

68. Cao, Y., Kementchedjhieva, Y., Cui, R., Karamolegkou, A., Zhou, L., Dare, M., Donatelli, L. and Hershcovich, D. Cultural adaptation of recipes. *Transactions of the Association for Computational Linguistics*, 12:80–99, 2024. doi:10.1162/tacl_a_00634.

69. Sahin, O. and Aytekin-Sahin, G. Open-source multi-objective optimization software for menu planning. *Expert Systems with Applications*, 252:124213, 2024.

70. Carbonell, J. and Goldstein, J. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336, 1998.

71. Menichetti, G., Ravandi, B., Mozaffarian, D. and Barabási, A. Machine learning prediction of the degree of food processing. *Nature Communications*, 14(1):2312, 2023. doi:10.1038/s41467-023-37457-1.

72. Atkinson, F. S., Brand-Miller, J. C., Foster-Powell, K., Buyken, A. E. and Goletzke, J. International tables of glycemic index and glycemic load values 2021: a systematic review. *American Journal of Clinical Nutrition*, 114(5):1625–1632, 2021.

73. Pappalardo, F., Russo, G., Tshinanu, F. M. and Viceconti, M. In silico clinical trials: concepts and early adoptions. *Briefings in Bioinformatics*, 20(5):1699–1708, 2019. doi:10.1093/bib/bby043.

74. Kovatchev, B. P., Breton, M., Dalla Man, C. and Cobelli, C. In silico preclinical trials: a proof of concept in closed-loop control of type 1 diabetes. *Journal of Diabetes Science and Technology*, 3(1):44–55, 2009. doi:10.1177/193229680900300106.

75. Dalla Man, C., Micheletto, F., Lv, D., Breton, M., Kovatchev, B. and Cobelli, C. The UVA/PADOVA type 1 diabetes simulator: new features. *Journal of Diabetes Science and Technology*, 8(1):26–34, 2014. doi:10.1177/1932296813514502.

76. Walonoski, J., Kramer, M., Nichols, J., et al. Synthea: an approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association*, 25(3):230–238, 2018.

77. Franklin, J. M., Schneeweiss, S., Polinski, J. M. and Rassen, J. A. Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases. *Computational Statistics & Data Analysis*, 72:219–226, 2014.

78. Kalnay, E. Atmospheric Modeling, Data Assimilation and Predictability. Cambridge University Press, Cambridge, 2003.

79. Hall, K. D., Sacks, G., Chandramohan, D., Chow, C. C., Wang, Y. C., Gortmaker, S. L. and Swinburn, B. A. Quantification of the effect of energy imbalance on bodyweight. *The Lancet*, 378(9793):826–837, 2011.

80. Look AHEAD Research Group. Eight-year weight losses with an intensive lifestyle intervention: the Look AHEAD study. *Obesity*, 22(1):5–13, 2014.

81. Diabetes Prevention Program Research Group. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *New England Journal of Medicine*, 346(6):393–403, 2002.

82. Wing, R. R., Lang, W., Wadden, T. A., Safford, M., Knowler, W. C., Bertoni, A. G., et al. Benefits of modest weight loss in improving cardiovascular risk factors in overweight and obese individuals with type 2 diabetes. *Diabetes Care*, 34(7):1481–1486, 2011.

83. European Federation of Clinical Chemistry and Laboratory Medicine. EFLM Biological Variation Database. https://biologicalvariation.eu, 2026.